

Sales Data Analysis

1. Project overview

This project analyzes ride booking and customer behavior using a dataset of 1,00,000 ride records collected. The dashboard focuses on understanding booking trends, cancellation patterns, ride performance, customer and driver ratings, vehicle-type demand, and revenue distribution. Using PostgreSQL, Python, and Power BI, the project uncovers insights into successful and cancelled rides, peak booking periods, average ride distances, booking values, and operational efficiency. The goal of this project is to help businesses make data-driven decisions to improve customer experience, optimize ride operations, and increase overall service performance.

2. Dataset Summary

- Rows: 1,00,000
- Columns: 20
- Key Features:
 - Ride booking details (Date, Time, Booking_ID, Booking_Status, Customer_ID)
 - Ride and vehicle information (Vehicle_Type, Pickup_Location, Drop_Location, Ride_Distance, Booking_Value)
 - Operational performance metrics (V_TAT, C_TAT, Incomplete_Rides, Incomplete_Rides_Reason)
 - Cancellation analysis (cancelled_Rides_by_Customer, cancelled_Rides_by_Driver)
 - Customer transaction insights (Payment_Method, Booking_Value)
 - Customer and driver feedback analysis (Driver_Ratings, Customer_Rating)
 - Demand and booking trend analysis across different vehicle categories and Bengaluru locations
 - Weekend and match-day ride surge analysis for operational and revenue insights
 - Missing Data: Missing values present in V_TAT, C_TAT, Booking_Value, Ride_Distance, Driver_Ratings, Customer_Rating, and cancellation/incomplete ride reason columns due to unsuccessful or incomplete bookings.

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas` & `numpy`
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Date	Time	Booking ID	Booking Status	Customer ID	Vehicle Type	Pickup Location	Drop Location	AvgVTAT	Avg CTAT	Cancelled Rides by Customer	Reason for cancelling by Customer	Cancelled Rides by Driver	Reasc cancelli
count	100000	100000	100000	100000	100000	100000	100000	100000	62000.000000	62000.000000	100000	7000	100000	
unique	31	52842	99478	5	60377	7	50	50	NaN	NaN	2	5	2	
top	10/5/2026	16:55:18	CNR9151228	Success	CUST62437	eBike	Banashankari	Nagawara	NaN	NaN	No	Driver is not moving towards pickup location	No	The cust coughin
freq	7543	11	2	62000	8	14526	2116	2134	NaN	NaN	93000	1606	82000	
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	6.992095	34.932942	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2.889683	14.418537	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2.000000	10.000000	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	4.500000	22.400000	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	7.000000	34.900000	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	9.500000	47.400000	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	12.000000	60.000000	NaN	NaN	NaN	

Reason for cancelling by Driver	Incomplete Rides	Incomplete Rides Reason	Booking Value	Ride Distance	Driver Ratings	Customer Rating
18000	100000	5000	62000.000000	62000.000000	62000.000000	62000.000000
4	2	3	NaN	NaN	NaN	NaN
The customer was coughing/sick	No	Other Issue	NaN	NaN	NaN	NaN
4552	95000	1701	NaN	NaN	NaN	NaN
NaN	NaN	NaN	510.733626	18.261066	4.249394	4.252773
NaN	NaN	NaN	363.366867	9.641512	0.435433	0.434092
NaN	NaN	NaN	80.000000	1.500000	3.500000	3.500000
NaN	NaN	NaN	258.927500	9.900000	3.900000	3.900000
NaN	NaN	NaN	427.870000	18.300000	4.300000	4.300000
NaN	NaN	NaN	662.097500	26.600000	4.600000	4.600000
NaN	NaN	NaN	3457.160000	35.000000	5.000000	5.000000

- Missing Data Handling:** Checked for null values and imputed missing values in the `V_TAT`, `C_TAT`, `Booking_Value`, `Ride_Distance`, `Driver_Ratings`, `Customer_Rating`, and `cancellation/incomplete ride reason` columns using the median rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
 - Created **payment_methods** column using random choice of numpy
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

1. **Successful Bookings Analysis** – Retrieved all rides that were completed successfully to analyze completed booking trends and ride performance.

	date text	time text	booking_id text	booking_status text	customer_id text	vehicle_type text	pickup_location text	drop_location text	avg_vat double precision	avg_ctat double precision	cancelled_rides_by_ci text
1	11/5/20...	15:05:...	CNR41882...	Success	CUST83407	Prime SUV	Wilson Garden	Hoodi	7.3	27.6	No
2	4/5/2026	19:37:...	CNR27883...	Success	CUST43885	eBike	Cunningham Road	Sahakar Nagar	4.5	32.1	No
3	2/5/2026	18:38:...	CNR98032...	Success	CUST22109	Prime Plus	Chickpet	JP Nagar	4.7	40.8	No
4	27/5/20...	9:46:23	CNR31606...	Success	CUST87282	eBike	KR Puram	Devanahalli	8.8	37.2	No
5	23/5/20...	21:22:...	CNR58727...	Success	CUST53265	Prime Plus	Indiranagar	Domlur	6.1	32.8	No
6	1/5/2026	4:08:56	CNR28265...	Success	CUST47379	eBike	Begur	JP Nagar	2.6	17.3	No
7	31/5/20...	18:52:...	CNR80212...	Success	CUST63706	Prime Sedan	BTM Layout	Brookefield	8.2	43.2	No
8	7/5/2026	8:02:08	CNR25126...	Success	CUST66779	Bike	Majestic	Brookefield	6.7	43.4	No
9	6/5/2026	7:40:13	CNR45604...	Success	CUST34456	Mini	Malleshwaram	BTM Layout	2.9	35.4	No
10	17/5/20...	6:06:20	CNR39069...	Success	CUST23765	Bike	Anekal	Airport Road	5.2	49	No
11	10/5/20...	15:54:...	CNR20046...	Success	CUST71814	Prime Plus	Cunningham Road	Wilson Garden	11.6	26	No
12	15/5/20...	7:54:50	CNR10220...	Success	CUST35778	eBike	Anekal	Kalyan Nagar	2.5	56.3	No

2. **Average Ride Distance by Vehicle Type** – Calculated the average ride distance for each vehicle category to understand travel patterns across different vehicle types.

	vehicle_type text	avg_distance numeric
1	eBike	18.23
2	Auto	18.35
3	Bike	18.22
4	Prime Sedan	18.37
5	Mini	18.20
6	Prime Plus	18.25
7	Prime SUV	18.30

3. **Customer Cancellation Analysis** – Measured the total number of rides cancelled by customers to evaluate customer cancellation behavior.

	cancelled_by_customer bigint
1	7000

4. **Top 5 Customers by Ride Volume** – Identified the top 5 customers with the highest number of bookings to analyze frequent user activity.

	customer_id text	total_rides bigint
1	CUST10181	8
2	CUST62437	8
3	CUST72025	7
4	CUST61031	7
5	CUST25147	7

5. **Driver Cancellation Reason Analysis** – Calculated the number of rides cancelled by drivers due to personal and car-related issues to identify operational challenges.

	cancelled_rides_by_driver bigint
1	4545

6. **Driver Rating Analysis for Prime Sedan** – Determined the maximum and minimum driver ratings for Prime Sedan rides to evaluate driver performance quality.

	max_rating double precision	min_rating double precision
1	5	3.5

7. **UPI Payment Usage Analysis** – Retrieved all rides where customers used UPI as the payment method to analyze digital payment preferences.

	reason_for_cancelling_by_driver text	incomplete_rides text	incomplete_rides_reason text	booking_value double precision	ride_distance double precision	driver_ratings double precision	customer_rating double precision	payment_method text
1	Not Applicable	No	Not Applicable	355.67	18	4.8	4.7	UPI
2	Personal & Car related issues	No	Not Applicable	427.87	18.3	4.249393548387097	4.2527725806451615	UPI
3	Not Applicable	No	Not Applicable	986.36	28.9	3.6	3.6	UPI
4	Personal & Car related issues	No	Not Applicable	427.87	18.3	4.249393548387097	4.2527725806451615	UPI
5	Not Applicable	No	Not Applicable	106.82	2.5	4.8	4.6	UPI
6	The customer was coughing/sick	No	Not Applicable	427.87	18.3	4.249393548387097	4.2527725806451615	UPI
7	Not Applicable	No	Not Applicable	785.08	13.2	4.4	4.3	UPI
8	Not Applicable	No	Not Applicable	427.87	18.3	4.249393548387097	4.2527725806451615	UPI

8. **Customer Rating by Vehicle Type** – Calculated the average customer rating for each vehicle category to measure customer satisfaction levels.

	vehicle_type text	avg_customer_rating numeric
1	eBike	4.25
2	Auto	4.25
3	Bike	4.25
4	Prime Sedan	4.25
5	Mini	4.25
6	Prime Plus	4.25
7	Prime SUV	4.26

9. **Successful Booking Revenue Analysis** – Calculated the total booking value generated from successfully completed rides to evaluate total revenue performance.

	total_successful_ride_value numeric
1	31665484.83

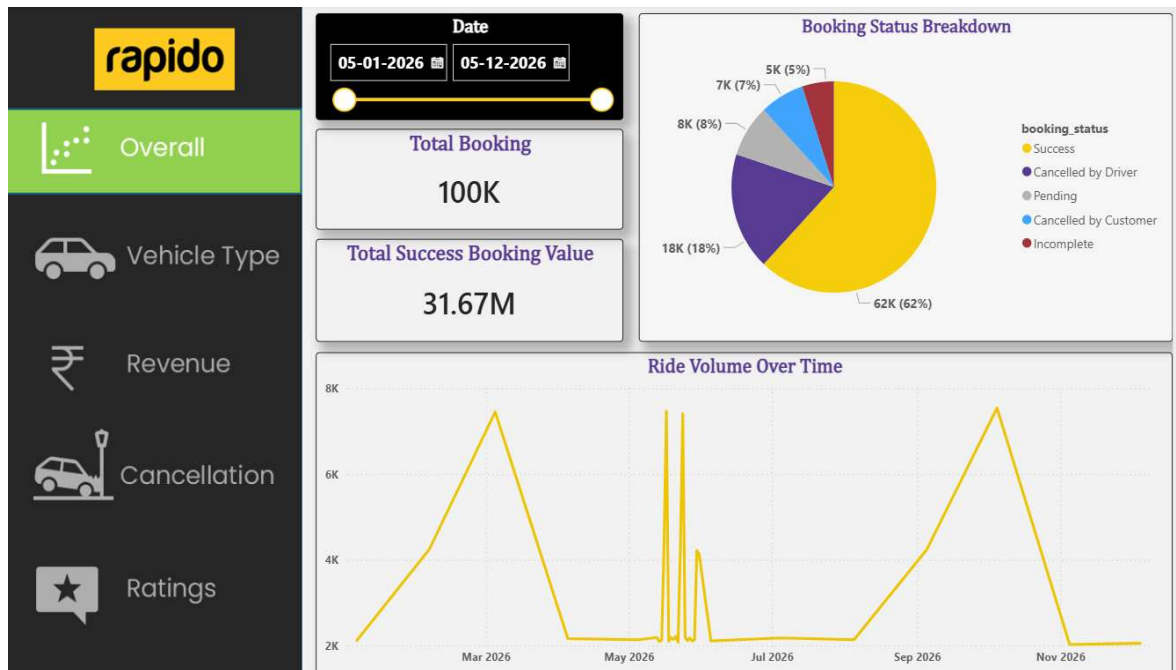
10. **Incomplete Ride Analysis** – Retrieved all incomplete rides along with their reasons to identify major causes of ride incompletion.

	booking_id text	incomplete_rides_reason text
1	CNR39227...	Customer Demand
2	CNR50761...	Customer Demand
3	CNR87963...	Customer Demand
4	CNR88909...	Customer Demand
5	CNR66812...	Vehicle Breakdown
6	CNR49425...	Other Issue
7	CNR70027...	Vehicle Breakdown
8	CNR82315...	Vehicle Breakdown
9	CNR58257...	Vehicle Breakdown
10	CNR72971...	Customer Demand
11	CNR69488...	Customer Demand
12	CNR20234	Vehicle Breakdown

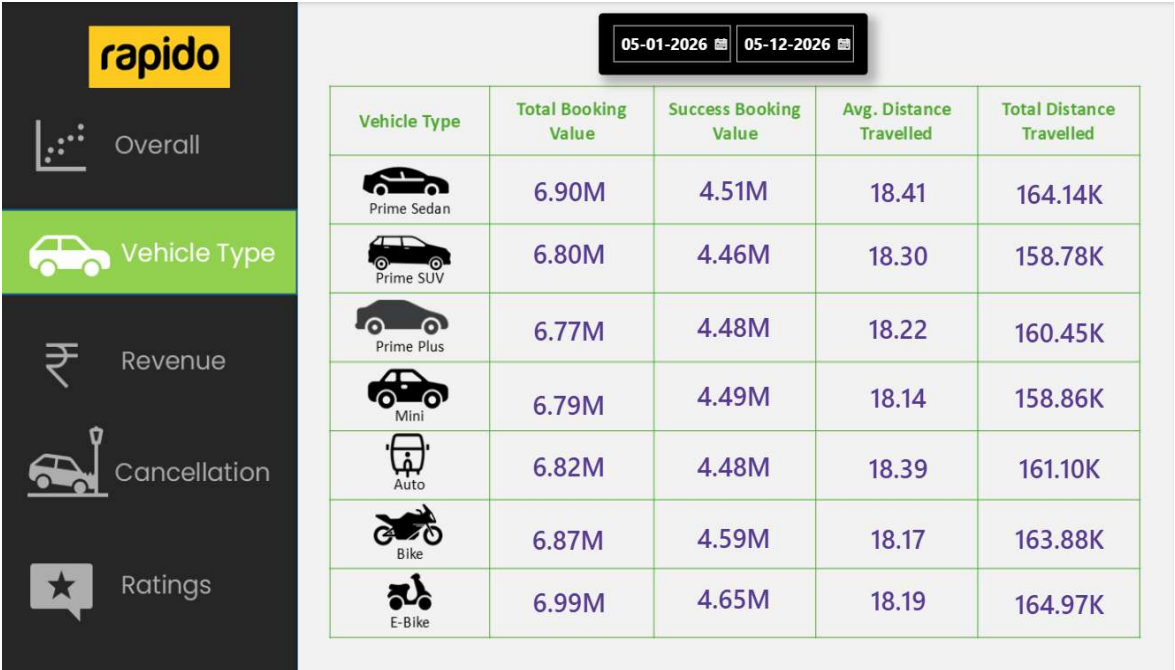
5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

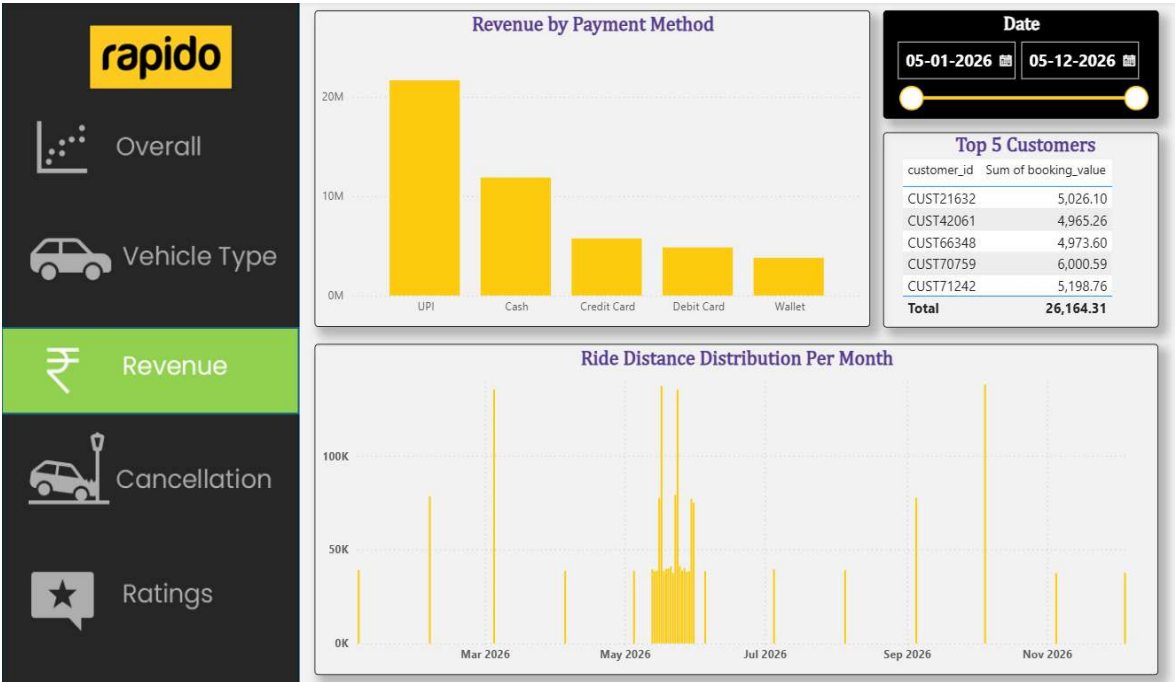
- Overall :



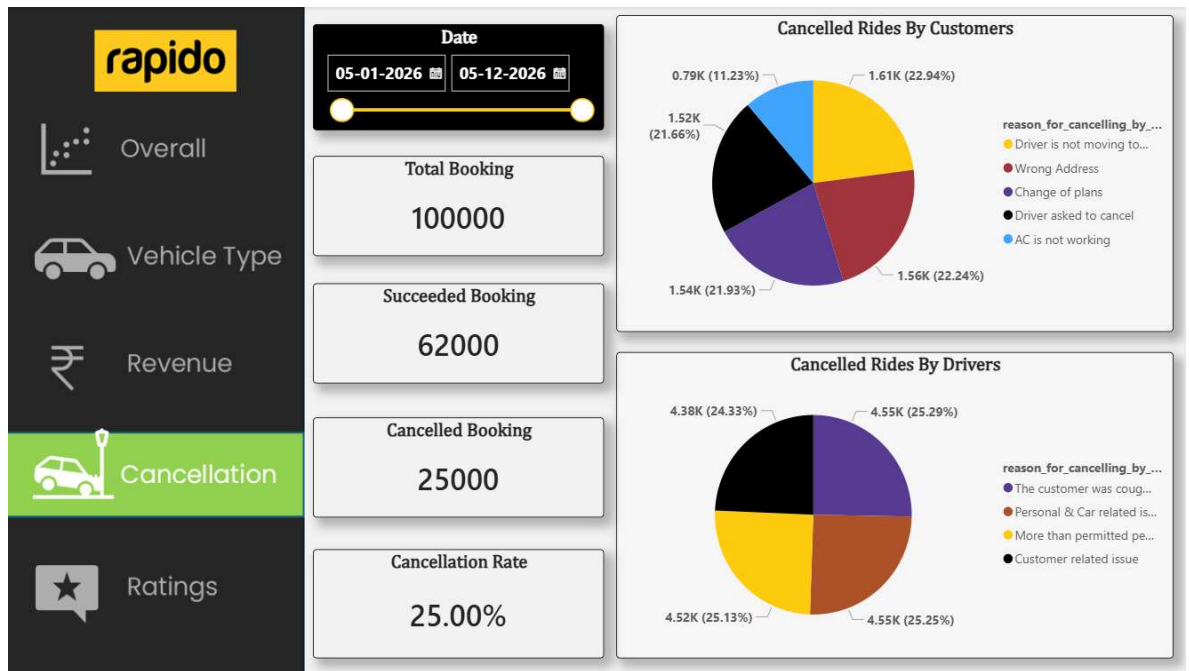
• Vehicle Type :



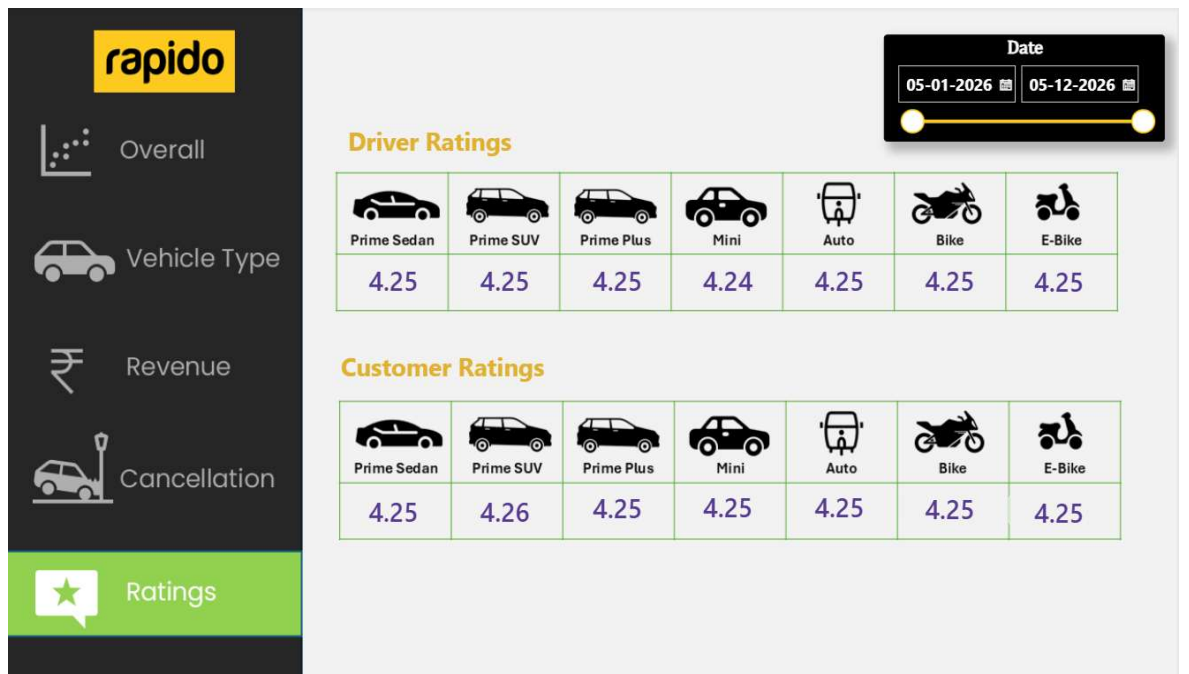
• Revenue :



- Cancellation :



- Ratings :



6. Business Recommendations

- ❖ **Reduce Ride Cancellation Rates** – Implement better driver allocation, stricter cancellation policies, and real-time tracking to minimize customer and driver cancellations.
- ❖ **Improve Customer Satisfaction** – Monitor low-rated rides and provide driver training programs to enhance ride quality and improve customer ratings.
- ❖ **Focus on High-Demand Locations** – Deploy more drivers in high-booking Bengaluru areas to reduce waiting time and improve ride availability.
- ❖ **Increase Weekend and Match-Day Revenue** – Apply surge pricing and targeted promotional campaigns during high-demand periods to maximize booking value and revenue.
- ❖ **Improve Operational Reliability** – Reduce incomplete rides and vehicle breakdowns through regular vehicle maintenance and operational monitoring.